

Lab Exercise: Schematics with ESP32

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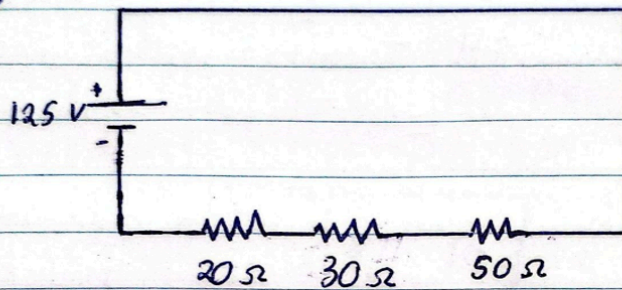
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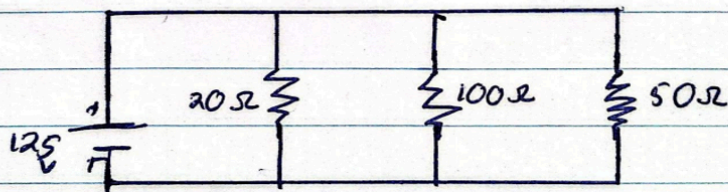
Lab Exercise.

Part A

a)



b)



① Calculate the total resistance in ohms.

a) series

$$20 + 30 + 50$$

$$= \underline{100 \Omega}$$

b) Parallel

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{30} + \frac{1}{50} = \frac{15 + 10 + 6}{300}$$

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{5 + 1 + 2}{100}$$

$$R = \frac{100}{8} = \underline{12.5 \Omega}$$

2. Calculate the total current from the power supply.

$$c) I = \frac{V}{R}$$

$$= \frac{12.5}{100} = \underline{1.25 A}$$

$$P = VI$$

$$b \quad \frac{125}{12.5} = \underline{10A}$$

3. Calculate the current through each resistor.

a) = current in a series circuit is same everywhere.

$$\frac{8}{100} = \frac{8}{2+1+2} = \frac{1}{100} + \frac{1}{100} + \frac{1}{100} = \frac{1}{100}$$

$$I_T = I_1 = I_2 = I_3 = 1.25A$$

$$R_{2.8.1} = \frac{100}{8} = 12.5$$

b) Parallel

1) ~~20 20 20~~

$$I = \frac{V}{R} = \frac{125}{20} = \underline{6.25A}$$

$$A_{2.8.1} = \frac{100}{100}$$

$$ii) I = \frac{V}{R} = \frac{125}{100} = 1.25A$$

$$iii) \frac{125}{50} = 2.5A$$

4 Calculate the voltage drop through each resistor.

$$V = IR$$

a) series.

i) 20Ω

$$V = 1.25 \times 20 = 25V$$

$$ii) 1.25 \times 30$$

$$= 37.5V$$

$$iii) 1.25 \times 50$$

$$= 62.5V$$

b) parallel

in a parallel circuit voltage drops
equal

$$V_1 = V_2 = V_3 = \underline{125\text{ V}}$$

5. Calculate the power dissipated through
each resistor. $P = VI$

a) series

$$P \text{ @ } 20\Omega$$

$$P = 125 \times 1.25 = 156.25\text{ W}$$

$$= \underline{31.25\text{ W}}$$

$$11) 30\Omega \quad V = 125\text{ V}$$

$$= 30 \times 1.25 = 37.5\text{ W}$$

$$= \underline{46.875\text{ W}}$$

$$\text{iii) } 50 \Omega$$

$$62.5 \times 1.25$$

$$= 78.125 \text{ W}$$

b) Parallel

$$\text{① } P = VI$$

$$\text{i) } 20 \Omega$$

$$P = 125 \times 6.25$$

$$= 78.125 \text{ W}$$

$$\text{ii) } 100 \Omega$$

$$P = 125 \times 1.25$$

$$= \underline{156.25 \text{ W}}$$

$$\text{iii) } 50 \Omega$$

$$P = 125 \times 2.5$$

$$= \underline{312.5 \text{ W}}$$

Ref	Component	Value/Part	Quantity
U1	ESP32 module	ESP32-WROOM-3 2E-N8	1
U2	Temp/humidity sensor	DHT22	1
U3	3.3 V regulator	NCP1117LPST33T 3G	1
USB1	USB Type-C Connector	USB3.1 TYPE-C 16P	1
R1, R2	CC pull-down resistors	5.1 kΩ	2
R3	EN pull-up resistor	10 kΩ	1
R4	DATA pull-up resistor	10 kΩ	1
C1, C2	Regulator input/output capacitors	10 μF	2
C3	EN capacitor	100 nF	1
R4	DATA pull-up resistor	10 kΩ	1
C1, C2	Regulator input/output capacitors	10 μF	2
C3	EN capacitor	100 nF	1
C4, C5	Decoupling capacitors	100 nF	2

Table 1: List of all components

9:44 PM | hvo-xbao-amp

Benedict Kirui (You, presenting) Stop presenting

People

Add people

Contributors 6

- Benedict Kirui (You) Meeting host
- Benedict Kirui Your presentation
- Emmanuel Ronoh
- Kelly Wangwe
- Leon Muema
- Ted Kamau

3 others

4 Calculate the voltage drop through each resistor using voltage divider

$V = IR$

a) 50×165

20Ω

$V = 1.35 \times 20 = 27 \times 20 = 4$

$= 85 \text{ V}$

$10 \times 1.35 \times 3.0$

$= 37.5 \text{ V}$

Figure 1: Proof of discussion